Resampling serves several purposes in algorithmic trading:

- Converting high-frequency data to lower frequencies for strategy development
- Creating multi-timeframe analyses (e.g., comparing daily and weekly trends)
- Reducing noise in very short-term data

Building a Complete Data Pipeline

Let's put everything together to create a complete data pipeline—a reusable process for fetching, cleaning, and preparing financial data for analysis:

```python
import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from datetime import datetime, timedelta

def get_clean_data(ticker, start_date, end_date=None, interval='1d'):
    """
    Fetch and clean financial data for a given ticker.

    Parameters:
    ticker (str): The ticker symbol
    start_date (str): Start date in 'YYYY-MM-DD' format
    end_date (str, optional): End date in 'YYYY-MM-DD' format, defaults to today
    interval (str, optional): Data frequency ('1d', '1wk', '1mo')

    Returns:
    pd.DataFrame: Clean, processed data ready for analysis
    """
    # Set default end date to today if not provided
    if end_date is None:
        end_date = datetime.today().strftime('%Y-%m-%d')
```